

- 9 (a) any value greater than, or equal to, $5\text{ k}\Omega$ B1 [1]
- (b) (i) 'positive' shown in correct position B1 [1]
- (ii) $V^+ = (500/2200) \times 4.5$
 $\approx 1\text{ V}$
 $V^- > V^+$ so output is negative
 green LED on, (red LED off)
(allow full ecf of incorrect value of V^+) B1
 M1
 A1 [3]
- (iii) *either* V^+ increases *or* $V^+ > V^-$
 green LED off, red LED on M1
 A1 [2]
- 10 quartz/piezo-electric crystal B1
 p.d. across crystal causes *either* centres of (+) and (–) charge to move
or crystal to change shape B1
 alternating p.d. (in ultrasound frequency range) causes crystal to vibrate B1
 crystal cut to produce resonance B1
 when crystal made to vibrate by ultrasound wave M1
 alternating p.d. produced across the crystal A1 [6]
- 11 (a) sharpness: ease with which edges of structures can be seen B1